

# CYBOX-S: FIELD GUIDE & PRACTICAL WORKFLOW REFERENCE

High-Speed Offline Penetration Testing Worksheet · eJPTv2 / eCPPT Practical Companion

## What is CYBOX-S?

CYBOX-S is a fast, keyboard-driven terminal worksheet and operational cockpit designed to eliminate exam cognitive overload. It provides an offline state machine for host discovery, port tracking, credential reuse, syntax cheatsheets, and question proofs. **It does not run autonomous exploits or rely on AI.** You retain 100% human control while CYBOX-S manages your operational memory.

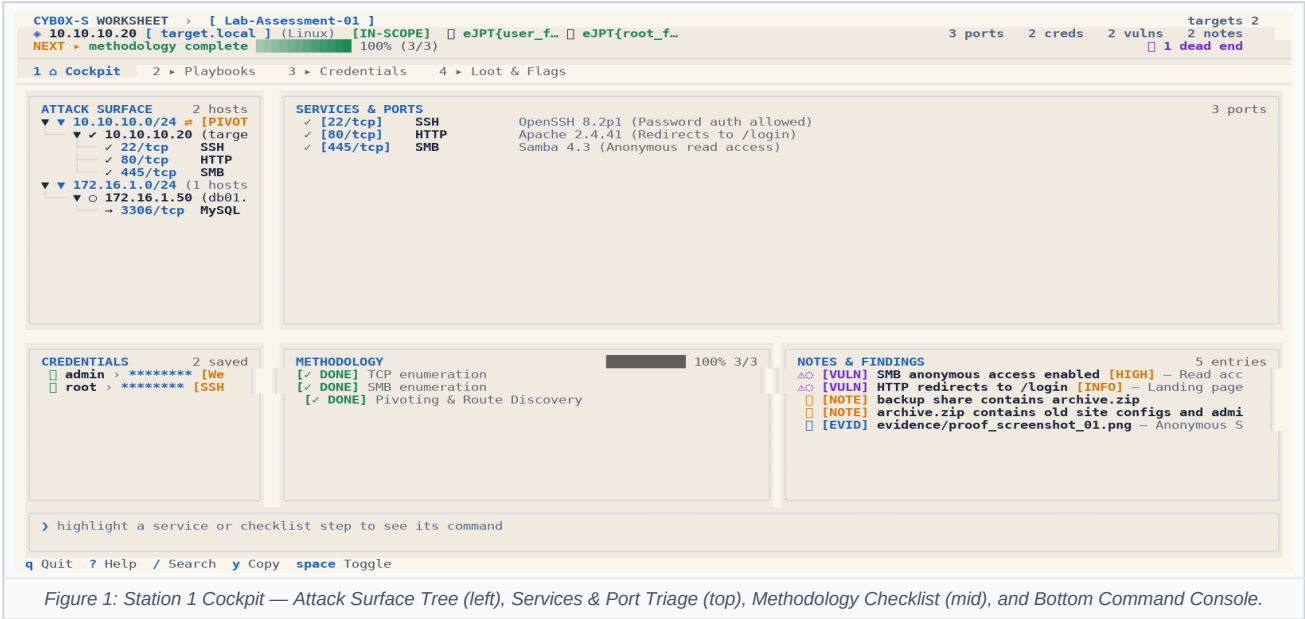
## INE eJPT / eCPPT Exam Compliance

- **100% Passive & Offline:** Runs entirely on localhost via local SQLite database.
- **Zero Autonomous Action:** Commands are copied to your clipboard; YOU execute them.
- **Zero Cloud / AI Dependencies:** No external API calls, leaks, or prohibited LLMs.
- **Permitted Personal Notes:** Functions strictly as a local worksheet and syntax lookup.

## 1. The Four Operational Stations (Switch with Keys 1, 2, 3, 4)

| Station   | Name & Purpose               | What You See & Do Here                                                                                                                                                           | Hotkey |
|-----------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Station 1 | Cockpit (Workbench)          | Targets tree, open ports, service triage status, methodology checklist, scratchpad notes, and interactive command console.                                                       | [1]    |
| Station 2 | Playbook Browser             | Browse full tactical playbooks (eJPT workflow, Web App OWASP, Active Directory, Pivoting, Linux & Windows PrivEsc) with instant copy.                                            | [2]    |
| Station 3 | Credential Matrix            | 2D grid of discovered credentials (rows) × target services (cols). Press [Enter] to compile & copy spray commands; [Space] to cycle verification state.                          | [3]    |
| Station 4 | Exam Proofs, Loot & Evidence | Track assessment question answers (via :q), user/root flags, disk-backed loot browser (with [Space] preview), clipboard screenshot proofing (:paste-ev / [V]), and Failure Logs. | [4]    |

## 2. Station 1 Cockpit Architecture & Real-Time Workspace



3. Complete Keyboard Shortcuts Cheatsheet ("Muscle Memory Map")

CYBOX-S is 100% operational from the keyboard. Single-letter keys trigger immediate actions unless an input modal is active:

| Key           | Category         | Action & Operational Behavior                                                                                                                                                                                                                                                                                                                         |
|---------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1, 2, 3, 4    | Station          | Switch immediately to Station 1 (Cockpit), 2 (Playbooks), 3 (Creds Matrix), or 4 (Loot & Flags).                                                                                                                                                                                                                                                      |
| I             | Ingestion        | <b>Import Scan &amp; Enum:</b> Interactive review modal for Nmap (-oX, -oN, -oG) and Web Enum (FFUF, Ferox, Gobuster). Toggle items before committing; archives evidence to scans/ or enum/.                                                                                                                                                          |
| W             | Workspace        | <b>Workspace Manager:</b> Switch between lab workspaces or scaffold a new per-lab directory with standard folders (scans/, enum/, screenshots/, loot/).                                                                                                                                                                                               |
| w             | Navigation       | <b>Cycle Panels:</b> Sequentially rotate focus across all panels in Cockpit without Tab.                                                                                                                                                                                                                                                              |
| h / l         | Navigation       | <b>Column Jump:</b> Press h for Sidebar (left); press l for Workbench (right).                                                                                                                                                                                                                                                                        |
| j / k         | Navigation       | <b>Vim Navigation:</b> Move highlight down (j) or up (k) in the active list or tree.                                                                                                                                                                                                                                                                  |
| [ / ]         | Targeting        | <b>Target Switcher:</b> Switch immediately to previous [ or next ] target machine.                                                                                                                                                                                                                                                                    |
| b             | Display          | <b>Sidebar Toggle:</b> Collapse sidebar to give 100% width to Workbench (press b again to restore).                                                                                                                                                                                                                                                   |
| : (colon)     | Console          | <b>Focus Command Bar:</b> Instantly activates the bottom console input ready to type.                                                                                                                                                                                                                                                                 |
| Esc           | Console          | <b>Dismiss / Blur:</b> Clears command input and returns focus directly to the active workbench.                                                                                                                                                                                                                                                       |
| Space         | Triage / Preview | <ul style="list-style-type: none"><li>• <b>On Service:</b> Cycle triage status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED].</li><li>• <b>In Station 3 Matrix:</b> Cycle test state: [UNTESTED] → [VALID] → [PWN3D] → [INVALID].</li><li>• <b>In Station 4 Loot:</b> Open <b>Loot Preview Modal</b> to inspect hashes, configs, or keys.</li></ul> |
| , / .         | Recipe Carousel  | <b>Tool Carousel:</b> When a service is highlighted, press . (next) or , (prev) to cycle alternative tool recipes in the console (e.g. feroxbuster → gobuster → nikto → curl).                                                                                                                                                                        |
| Enter         | Execute / Copy   | <ul style="list-style-type: none"><li>• <b>On Guidance / Service:</b> Copies the previewed tool command to clipboard.</li><li>• <b>In Station 3 Matrix:</b> Compiles &amp; copies ready-to-run spray command for credential &amp; port.</li><li>• <b>In Station 4 Loot:</b> Copies relative file path to clipboard (e.g. loot/users.txt).</li></ul>   |
| v             | Evidence         | <b>Paste Screenshot:</b> In Station 4, saves clipboard image directly to screenshots/ and links as Evidence.                                                                                                                                                                                                                                          |
| y             | Quick Copy       | Copies selected entity's primary value (IP address, port, password, or checklist command).                                                                                                                                                                                                                                                            |
| z             | Layout           | <b>Zoom:</b> Maximize the focused panel to full-screen; press z again to restore normal layout.                                                                                                                                                                                                                                                       |
| T             | Theme            | <b>Theme Picker:</b> Open visual theme modal with 10 vibrant palettes (press d to set default).                                                                                                                                                                                                                                                       |
| / or Ctrl+F   | Search           | <b>Global Fuzzy Search:</b> Search across targets, services, credentials, checklists, and notes.                                                                                                                                                                                                                                                      |
| r             | Reference        | <b>Quick Reference Modal:</b> Open offline practical cheatsheet for instant syntax lookup.                                                                                                                                                                                                                                                            |
| t / s / c / n | Fast Capture     | Modal dialogs to quickly add Target (t), Service (s), Credential (c), or Note (n).                                                                                                                                                                                                                                                                    |
| a             | Question Proof   | In Station 4: Open <b>Add Proof</b> modal to link question/item number to proof evidence.                                                                                                                                                                                                                                                             |

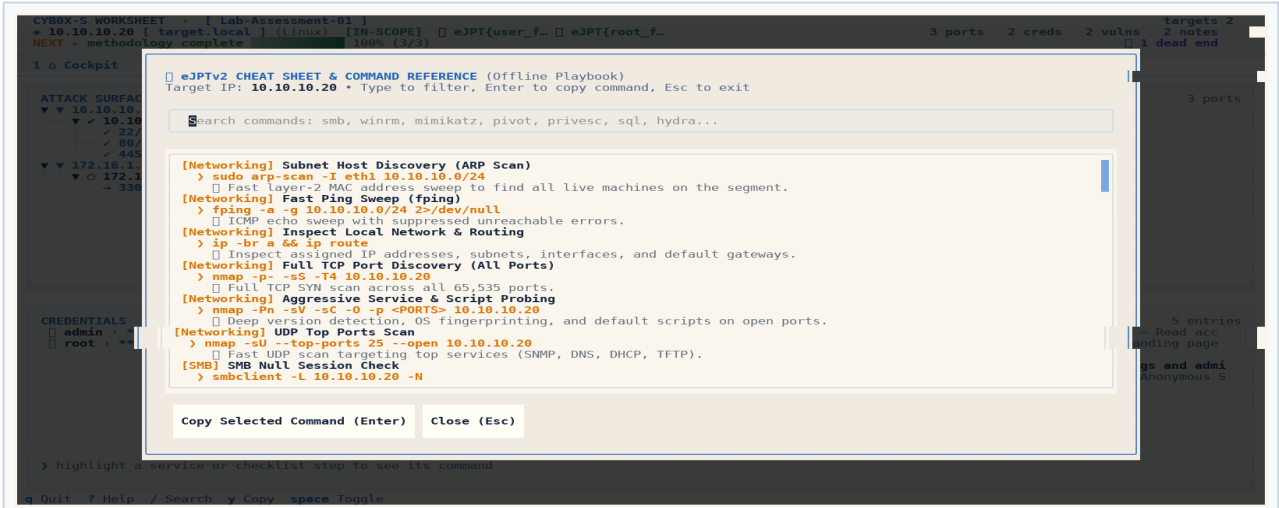


Figure 2: Practical Reference Cheatsheet Modal (Hotkey [r]) — Instant syntax lookup for Nmap, NetExec, Hydra, Chisel, and LinPEAS.

**PRO-TIP: THE ZERO-MOUSE KEYBOARD FLOW**

1. Highlight any port in **SERVICES & PORTS** with j / k.
2. Watch the bottom console display the exact syntax with the target IP already substituted.
3. Press . to cycle alternative tools (e.g., feroxbuster → gobuster → nikto → curl).
4. Press Enter to copy the command directly to your terminal clipboard, switch to your tmux/terminal pane, and run it!
5. Press Space to mark the port [CHECKED] or [DEAD-END].

4. Bottom Console Syntax & Wordlist Accelerators

| Console Command                                   | Description & Operational Effect                                                                                                                        | Example Syntax                                                                                                                               |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <code>:w &lt;alias&gt;</code>                     | <b>Wordlist Accelerator:</b> Instantly copies standard Kali / SecLists paths to clipboard.                                                              | <code>:w rockyou</code> <code>:w common</code> <code>:w medium</code><br><code>:w raft-d</code> <code>:w raft-f</code> <code>:w users</code> |
| <code>:import &lt;path&gt;</code>                 | <b>Scan &amp; Enum Ingestion:</b> Interactive review modal for Nmap XML/text or Web Enum (FFUF, Ferret, Gobuster). Saves raw output to scans/ or enum/. | <code>:import scans/nmap.xml</code><br><code>:import enum/ffuf.json</code>                                                                   |
| <code>:ws [name init]</code>                      | <b>Workspace Manager:</b> Switch active lab or scaffold standard folder layout.                                                                         | <code>:ws lab01</code> <code>:ws init lab02</code>                                                                                           |
| <code>:lhost / :lport</code>                      | <b>Attacker IP &amp; Port:</b> Auto-detects VPN IP (tun0) and substitutes into all playbooks.                                                           | <code>:lhost auto</code> <code>:lport 4444</code>                                                                                            |
| <code>:t &lt;ip&gt; [opts]</code>                 | Add target with optional host, OS, subnet, and pivot flag.                                                                                              | <code>:t 10.10.10.5 host linux</code><br><code>10.10.10.0/24 pivot</code>                                                                    |
| <code>:s &lt;port&gt; &lt;svc&gt;</code>          | Record newly discovered port & service on active target.                                                                                                | <code>:s 445/tcp smb</code> <code>:s 8080 http</code>                                                                                        |
| <code>:c &lt;user:pass&gt;</code>                 | Record discovered credential with optional service scope.                                                                                               | <code>:c admin:Secret123! SMB</code> <code>:c root:toor SSH</code>                                                                           |
| <code>:c crack &lt;id&gt; &lt;plain&gt;</code>    | <b>In-Place Cracking:</b> Upgrades captured hash to plaintext secret without duplicating records.                                                       | <code>:c crack 1 Password123</code>                                                                                                          |
| <code>:export wordlists</code>                    | <b>Spray Wordlist Export:</b> Generates deduplicated loot/users.txt and passwords.txt.                                                                  | <code>:export wordlists</code> <code>:export creds</code>                                                                                    |
| <code>:paste-ev / :ev latest</code>               | <b>Screenshot Proofing:</b> Saves clipboard image to screenshots/ or links newest OS capture.                                                           | <code>:paste-ev Root proof</code><br><code>:ev latest Question 14</code>                                                                     |
| <code>:q / :export</code>                         | Pin assessment question proof or export markdown evidence dossier.                                                                                      | <code>:q 7 /var/www/wp-config.php</code><br><code>:export exam</code> <code>:export report</code>                                            |
| <code>:uflag / :rflag &lt;hash&gt;</code>         | Record user flag or root/admin flag into Station 4 Loot Ledger.                                                                                         | <code>:uflag 7a9e...</code> <code>:rflag f04b...</code>                                                                                      |
| <code>:st &lt;where&gt; / :cl &lt;clue&gt;</code> | Record stuck point (rabbit hole) and breakthrough clue.                                                                                                 | <code>:st stuck on wp-login</code> <code>:cl found backup.zip</code>                                                                         |
| <code>:m &lt;template&gt;</code>                  | Instantiate standard checklist template (replace or append).                                                                                            | <code>:m ejpt</code> <code>:m web</code> <code>:m smb append</code>                                                                          |
| <code>:theme &lt;name&gt;</code>                  | Switch theme live (e.g. sugary, midnight, slate, caramel, cyber).                                                                                       | <code>:theme sugary</code> <code>:theme midnight</code>                                                                                      |

5. Step-by-Step Practical Exam Workflow ("The Battle Plan")

**PHASE 1: SCOPING & HOST DISCOVERY**

1. **Host Discovery:** Identify alive target IPs (nmap -sn 10.10.10.0/24 or arp-scan).

2. **Fast Add to CYBOX-S:** Type `:t 10.10.10.20` into console. Assign subnets: `:subnet 10.10.10.20 10.10.10.0/24`.

3. **Scope & Routing:** Tag dual-homed pivot machines with `:pivot 10.10.10.20 172.16.1.0/24` for automatic guidance.

4. **Import XML Option:** Mass scans can be imported directly: `cybox-s import /path/to/scan.xml`.

**PHASE 2: DETAILED PORT & SERVICE ENUMERATION**

1. **Full Port Sweep:** `nmap -sS -p- --min-rate 1000 <IP>`, then version scan: `nmap -sV -sC -p <ports> <IP>`.

2. **Record Open Services:** Type `:s 80 http`, `:s 445 smb`, `:s 3306 mysql` into the console.

3. **Inspect Tactical Guidance:** Highlight each service in SERVICES & PORTS. Press `.` to cycle tool recipes (whatweb → feroxbuster → gobuster → nikto → curl). Press **[Enter]** to copy.

4. **Service Triage with [Space]:** Cycle status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED].

**PHASE 3: WEB & LOW-HANGING FRUIT EXPLOITATION**

1. **Inspect Anonymous / Default Access:** Check anonymous FTP (`:s 21`), null SMB shares (`:s 445`), and SNMP strings (`:s 161`).

2. **Directory Fuzzing:** Type `:w common` or `:w medium` to copy the SecLists directory path. Paste directly into feroxbuster or gobuster.

3. **Cross-Filtering:** Highlighting a service in SERVICES & PORTS auto-scrolls the **Methodology** checklist to that service's testing steps!

**PHASE 4: CREDENTIAL SPRAYING & LATERAL MOVEMENT (STATION 3)**

1. **Record Every Credential:** Whenever you find creds (web configs, DB dumps, code), press **[c]** or type `:c user:pass scope`.

2. **Open Station 3 Matrix:** Press **[3]** to switch to the 2D Credential Spray Matrix.

3. **Instant Spray Generation:** Highlight any cell and press **[Enter]** to compile & copy ready-to-run spray commands (netexec, evil-winrm).

4. **Record Verification State:** Press **[Space]** on the cell to cycle: [UNTESTED] → [VALID] → [PWN3D] → [INVALID]. State persists in SQLite!

**PHASE 5: PRIVESC, PIVOTING & EVIDENCE CAPTURE (STATION 4)**

1. **Pivoting & Route Addition:** Tag pivot hosts with `:pivot` to see dynamic proxy guidance in the console bar.

2. **Privilege Escalation:** Switch to Station 2 (press **[2]**) and search Linux SUID, sudo -l, Cron jobs, or Windows tokens / services.

3. **Capture Question Evidence & Flags:** Log question proofs with `:q <num> <proof>` (or press **[a]** in Station 4). Record flags with `:uflag / :rflag`.

4. **Export Assessment Submission Bundle:** Run `:export exam` to write a clean Markdown report with all evidence, flags, and pivot paths!

**THE 20-MINUTE ROTATION RULE (EXAM GOLDEN RULE)**

If you have been analyzing a single service for more than 20 minutes without a new lead: **STOP**. Press **[Space]** on the service to mark it [DEFERRED]. Type `:stuck <service description>` to log where you are stuck, press **[j]** to rotate to another machine or port, and harvest low-hanging fruit elsewhere. 80% of exam failures occur because candidates spend 3 hours trapped down an unexploitable rabbit hole!

6. Station 3 Deep Dive: 2D Credential Spray Matrix

In multi-machine exams (like eJPTv2 / eCPPT), credential reuse across SSH, SMB, WinRM, and Web portals is the primary lateral movement technique. Station 3 prevents credential amnesia by organizing all discovered credentials (rows) against all discovered target services (columns) in an interactive 2D grid:

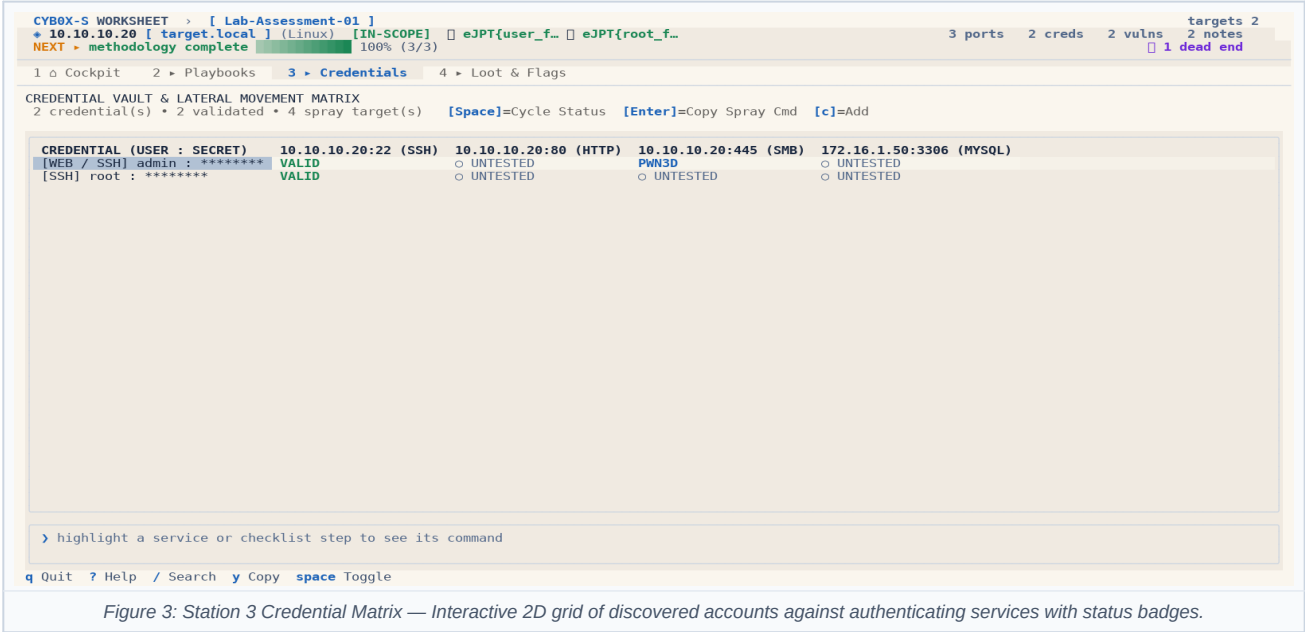


Figure 3: Station 3 Credential Matrix — Interactive 2D grid of discovered accounts against authenticating services with status badges.

| Action / Key           | Mechanism & Tactical Utility                                                                                                         | Persistence & Exam Value                                                           |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Arrow Keys / Vim       | Navigate freely across cells in the 2D grid.                                                                                         | Instant visual overview of which accounts were tested where.                       |
| [Space] (Cycle State)  | Cycles through: [UNTESTED] → [VALID] (Green) → [PWN3D] (Blue) → [INVALID] (Plum).                                                    | Persisted instantly to local SQLite database. Never guess if you tried a password. |
| [Enter] (Copy Command) | Automatically compiles and copies the exact command line to your clipboard (e.g. netexec smb 10.10.10.20 -u admin -p 'P@ssword123'). | Zero typos under stress; supports SSH, SMB, WinRM, RDP, MySQL, and MSSQL.          |
| [c] (Fast Add)         | Opens modal to record newly discovered credentials (or type :c user:pass [scope]).                                                   | Automatically expands the matrix with a new row across all existing targets.       |

LATERAL MOVEMENT WORKFLOW SUMMARY:

- Harvest:** Extract credentials from web app configs (wp-config.php, .env), DB dumps, or SAM registry hive.
- Input:** Press [c] in CYBOX-S to add the username and password.
- Matrix Spray:** Press [3] to open the grid. Highlight the intersection of the new credential with port 445 (SMB) or 22 (SSH).
- Execute:** Press [Enter] to copy the pre-built NetExec/SSH command, paste and run in your terminal.
- Record:** If admin/pwn3d, press [Space] twice to set [PWN3D]. You now have your next pivot point!

## 7. Station 4 Deep Dive: Question Proofs & Loot Ledger

Practical exams require concrete evidence: hashes, specific directory names, database records, usernames, or service versions. Station 4 provides a dedicated ledger to link lab findings directly to question numbers, plus a Failure Log to manage stuck points:



Figure 4: Station 4 Proofs & Loot Ledger — Question Proofs, User/Root Flags, and Failure / Rabbit Hole Logs.

| Module / Feature               | Hotkeys & Console Syntax                                                                     | Operational Purpose & Deliverable                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question Proofs                | Press <b>[a]</b> (modal) or <code>:q &lt;num&gt; &lt;proof&gt; [notes]</code>                | Links evidence directly to an exam question number (e.g. <code>:q 14 /etc/passwd</code> root hash). Prevents having to re-exploit targets at exam end. |
| Disk-Backed Loot Browser       | Press <b>[Space]</b> to preview<br>Press <b>[Enter]</b> to copy path                         | Scans loot/, screenshots/, enum/, and scans/. Preview text files, hashes, and dumps with instant copy.                                                 |
| Screenshot Proof Capture       | Press <b>[v]</b> or <code>:paste-ev [desc] / :ev latest</code>                               | Saves clipboard image directly to screenshots/ via Wayland/X11 or auto-attaches most recent screenshot.                                                |
| Loot & Flags Tracker           | <code>:uflag &lt;hash&gt;</code> (user flag)<br><code>:rflag &lt;hash&gt;</code> (root flag) | Captures proof hashes with timestamps and associated target IP. Safely stored in local SQLite database.                                                |
| Failure Log (Anti-Rabbit Hole) | <code>:st &lt;stuck_point&gt;</code><br><code>:cl &lt;breakthrough_clue&gt;</code>           | Records dead-ends and breakthroughs so you never repeat failed brute-force or injection attempts.                                                      |
| Markdown Submission Dossier    | Type <code>:export exam</code> or <code>:export report</code> in console                     | Generates an instant offline Markdown report summarizing all answered questions, flags, and credentials.                                               |

**THE OFFLINE SUBMISSION DOSSIER (:export exam):**

At any time, or before finalizing your exam answers in the INE portal, run `:export exam` in the bottom console. CYBOX-S exports a complete, self-contained Markdown file containing: (1) Target Inventory & Subnet Map, (2) Solved Questions 1–35 with exact evidence strings, (3) Captured User & Root Flags, (4) Discovered Credentials & Verification States, and (5) Audit Trail of all tested services. Keep this document open in your text editor while answering the 35 exam questions to achieve 100% submission confidence.

## 8. Station 2 Playbooks & Built-In Methodology Templates

Station 2 serves as an offline tactical encyclopedia. You can search playbooks for specific attack chains, inspect commands, and copy snippets with a single keystroke:

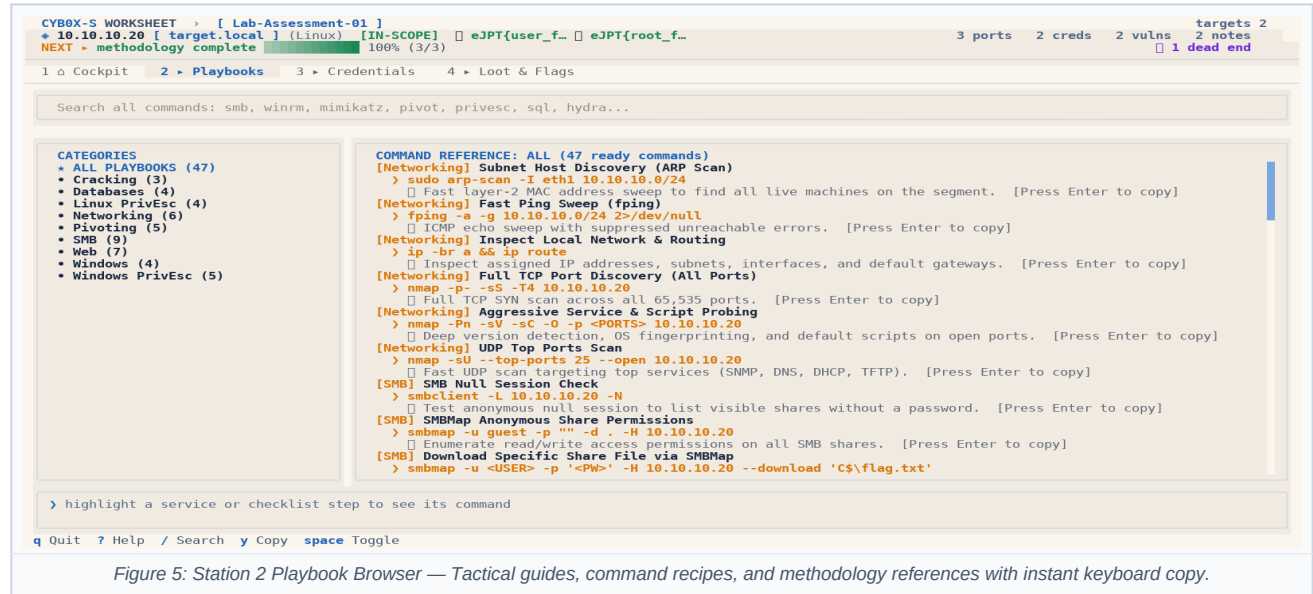


Figure 5: Station 2 Playbook Browser — Tactical guides, command recipes, and methodology references with instant keyboard copy.

| Template Alias     | Category         | Key Checklist Steps & Focus                                                                                                                 |
|--------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| :m ejpt            | eJPT Practical   | Scope recon → Host discovery → Full TCP scan → Service triage → Low-hanging fruit → Web fuzzing → Cred spray → Pivoting → PrivEsc.          |
| :m web             | OWASP Web App    | Technology profiling (whatweb) → Directory fuzzing (feroxbuster) → Parameter fuzzing → SQLi → Auth bypass → LFI/RFI → File upload.          |
| :m smb             | SMB & Windows    | Null session shares (smbclient/smbmap) → User enum (rpcclient) → Password policy → Anonymous signing audit → Vulnerability scan (MS17-010). |
| :m pivoting        | Network Pivoting | Dual-homed adapter discovery (ip a) → Internal route inspect → Proxychains & Chisel server setup → Port forward (socat) → Subnet sweep.     |
| :m privesc_linux   | Linux PrivEsc    | sudo -l commands → SUID/SGID binaries (GTFEBins) → Capabilities (getcap) → World-writable scripts & crontabs → Internal listening ports.    |
| :m privesc_windows | Windows PrivEsc  | whoami /priv (SeImpersonate) → Unquoted service paths → Modifiable service binaries → Stored credentials (cmdkey) → Registry backup.        |

EXAM START PROCEDURES:

1. Open terminal on your host or VM and launch: cyb0x-s

2. Pick your visual theme: press [T], select palette (e.g. Sugary, Midnight, Slate, or Cyber), press [d] to save as default.

3. Import initial Nmap scan: run cyb0x-s import /path/to/scan.xml or fast-add targets with :t <ip>.

4. **Offline Markdown Submission Bundle:** Run :export exam at any time to output a structured markdown report containing all recorded evidence, flags, credentials, and pivot paths.

5. **Database Safety:** Stored in ~/.local/share/cyb0x-s/worksheets.db. 100% local, persistent, zero network calls, and fully compliant with INE exam rules.